Math 372: Homework 08

Due Wednesday, March 25

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1. The density function for the standard normal distribution is the function

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

By computing first and second derivatives, show that ϕ is concave down for $|z| < 1$, concave up for $|z| > 1$, with inflection points at $z = \pm 1$. Carefully sketch the density curve.

Problem 2 (continuous r.v.)**.** The continuous random variable X has probability density function

$$f(x) = \begin{cases} 1/3 & : \quad 0 < x < 1 \\ 1/6 & : \quad 1 < x < 5 \\ 0 & : \quad \text{elsewhere} \end{cases}$$

(a) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$

(b) Compute $\mathrm{Var}(X)$

(c) Compute $\mathbb{E}[(X+1)^2]$

Problem 3. Let X be a random variable with Poisson distribution with parameter $\lambda > 0$. Using the formula $\mathbb{E}[X] = \sum_{k=0} k\mathbb{P}[X = k]$ show that $\mathbb{E}[X] = \lambda$. [Hint: use the fact that $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$.]

Problem 4 (continuous r.v.)**.** Let T be the waiting time, in hours, between successive purchases at a furniture store. Suppose that T has pdf

$$f_T(t) = Ce^{-\frac{t}{4}} \mathbf{1}_{[t \geq 0]} = \begin{cases} Ce^{-\frac{t}{4}} & : \quad t \geq 0 \\ 0 & : \quad t < 0 \end{cases}$$

where $C > 0$ is some number that you'll have to determine.

(a) Find a value of C so that $f_T(t)$ is a valid pdf.

(b) What is the probability that the time between purchases is greater than 6 minutes? [Note: 6 minutes $=$ 1/10 of an hour].

(c) Compute $\mathbb{E}[T]$.

Problem 5. The Pareto distribution is "power-law" distribution used for modeling things like income distribution and earthquake magnitudes. Specifically, given a random variable X, we say that the distribution of X is a member of the *Pareto family of distributions* if its cdt is

$$F(x) = \mathbb{P}[X \leq x] = \begin{cases} 1 - \left(\frac{b}{x}\right)^a & : \quad x \geq b \\ 0 & : \quad x < b \end{cases}$$

for some fixed positive parameters a and b.

(a) Derive the pdf of X. Make sure to specify the function on all parts of its domain: both when $x \geq b$ and when $x < b$.

(b) Show that the improper integral $\int_1^{\infty} \frac{1}{x^c} dx$ diverges to $+\infty$ when $c \geq 1$, but converges to $\frac{1}{c-1}$ when $c > 1$. *Hint:* recall that

$$\int_1^{\infty} \frac{1}{x^c} dx = \lim_{y \to \infty} \int_1^{y} \frac{1}{x^c} dx.$$

(c) What is $\mathbb{E}[X]$? Consider the cases with $a \leq 1$ and $a > 0$ separately.

Problem 6 (Exercise 4.4.61)**.** Data collected at Toronto Pearson International Airport suggests that an exponential distribution with mean value 2.725 is a good model for rainfall duration.

(a) What is the probability that the duration of a particular rainfall event at this location is at least 2 hours? At most 3 hours? Between 2 and 3 hoours?

(b) What is the probability that the raindfall duration exceeds the mean value by more than 2 standard deviations?

Problem 7 (Exercise 4.1.4)**.** Let X denote the vibratory stress (psi) on a wind turbine blade at a particular wind speed in a wind tunnel. The article "Blade Fatigue Life Assessment with Application to VAWTS" *(J. of Solar Energy Engr., 1982)* propose the Rayleigh distribution, with pdf

$$f(x) = \begin{cases} \frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}} & : \quad x > 0 \\ 0 & : \quad \text{otherwise} \end{cases}$$

as a model for the X distribution. (Here, $\theta > 0$ is a parameter.)

(a) Verify that $f(x)$ is a legitimate parameter for any $\theta > 0$.

(b) Suppose $\theta = 100$ (a value suggested in the article). What is the probability that X is at most 200? Less than 200? At least 200?

(c) What is the probability that X is between 100 and 200 (again assuming $\theta = 100$)?

(d) Give an expression for $\mathbb{P}[X \leq x]$.

Problem 8. Suppose X is a random variable with probability density function

$$f(x) = \begin{cases} 3x^2 & : \quad 0 < x < 1 \\ 0 & : \quad \text{otherwise} \end{cases}$$

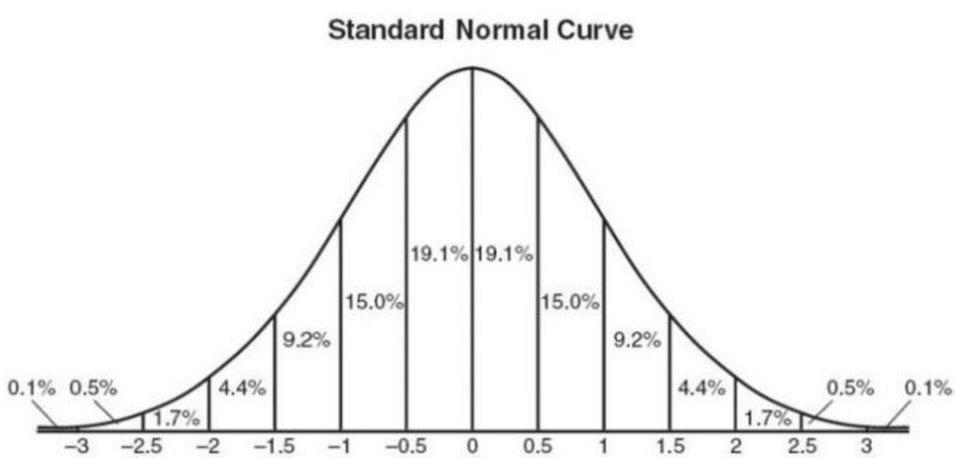
(a) Find the cumulative density function $F(x)$. *(Hint: don't forget to specify the value of $F(x)$ when $x < 0$ and $x > 0$.)*

(b) What is $\mathbb{P}[0.1 < X < 0.5]$? *(You don't need to simplify your answer)*

(c) What is $\mathrm{Var}(X)$?

(d) What is $\mathbb{E}\left[(X+3)^2\right]$?

For the remaining problems, the following curve of the density of a standard normal random variable Z may be helpful:



Problem 9 (Exercise 4.3.32)**.** Suppose the foce acting on a column that supports a building is normally distributed with mean value 15 kips and standard deviation 1.25 kips. Compute the following probabilities by standardizing and then inspecting the standard normal curve:

(a) $\mathbb{P}[X \leq 15]$

(b) $\mathbb{P}[X \leq 17.5]$

(c) $\mathbb{P}[X \geq 10]$

(d) $\mathbb{P}[14 \leq X \leq 18]$

(e) $\mathbb{P}[|X - 15| \leq 3]$

Problem 10 (Normal distribution)**.** Resting breathing rates for college-age students are approximately normally distributed with mean 12 and standard deviation 2.3 breaths per minute. What fraction of all college-age students have breathing rates in the following intervals?

(a) 9.7 to 14.3 breaths per minute

(b) 7.4 to 16.6 breaths per minute

(c) 9.7 to 16.6 breaths per minute

(d) Less than 5.1 or more than 18.9 breaths per minute

Problem 11 (Exercise 4.3.41)**.** The automatic opening device of a military cargo parachute has been designed to open when the parachute is 200 m above the ground. Suppose opening altitude actually has a normal distribution with mean value 200 m and standard deviation 30 m. Equipment damage will occur if the parachute opens at an altitude of less than 100 m. What is the probability that there is equipment damage to the payload of at least one of five independently dropped parachutes?